ALICE L WALTON SCHOOL OF MEDICINE

Vasculature and Lymphatics of the Heart

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Adapted from: Vasculature and Lymphatics of the Heart

Listen to this Brick

0:00 / 23:46 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Outline the origin, course, and distribution of the coronary arteries.
2. Discuss the typical variations in the arterial structures of the heart.
3. Outline the ac veins.
4. Describe the lymphatic drainage of the heart.

CASE CONNECTION

Over the past week, two patients, Alex and Sam, presented to the cardiac care unit with substernal chest pain. Each patient had a ruptured atherosclerotic plaque and a platelet plug in the proximal left anterior descending coronary artery. You note that the two patients had identical comorbidities, and both experienced 30 minutes of chest pain before identical treatment began. Alex had extensive myocardial ischemia and cell death and developed severe systolic dysfunction. Sam had lesser myocardial damage and had preserved left ventricular function.

Why might Sam have had a better outcome if everything else in the two patients' profiles and management was the same? Consider your answer as you read, and we'll revisit Alex and Sam at the end of the brick.

GO TO CONCLUSION ↓

What Are the Coronary Arteries? (LO#1)

How do we make sure this pump receives enough fuel to keep it going? The **coronary arteries** provide the heart with the oxygen and nutrients it needs.

Note: Revisit the Vascular Overview brick from KIFS as needed for reference.

Primary Coronary Arteries

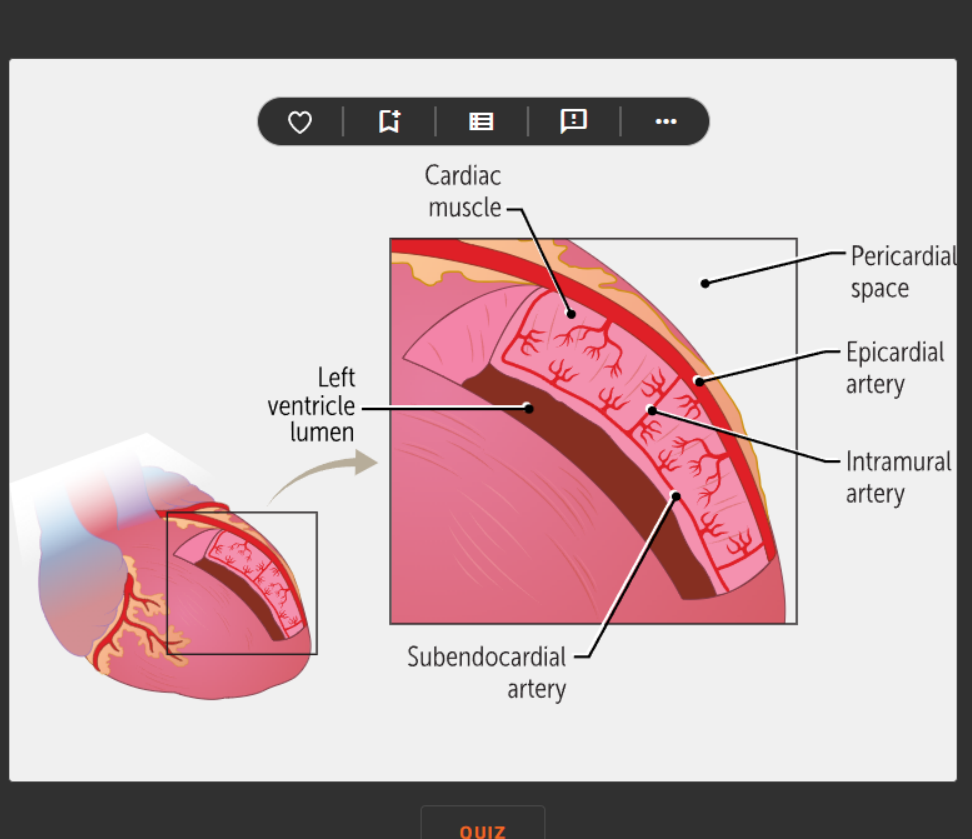
The coronary arteries originate from the aorta just distal to the aortic valve. The mechanical relationship between the coronary arteries and the aortic valve is ingenious. When the aortic valve is open during cardiac systole and blood pressure is at its highest (ventricular contraction), there is no flow in the coronary arteries. Compression of the coronary arteries during systole prevents the flow of blood into the coronary arteries.

However, when the left ventricle relaxes during diastole, the opening of the arteries is less occluded, and the backflow of blood causes the blood to enter the coronary arteries. From there, the blood then goes into a network of smaller vessels that supply the entire myocardium (the muscular layer of the heart).

CLINICAL CORRELATION

Because the coronary arteries emerge distal to the aortic valve, aortic stenosis (narrowing of the valve) will limit circulation to the periphery but also to the coronary arteries and can cause angina (ischemic chest pain from lack of oxygen).

The coronary arteries run on the external surface of the heart (epicardial arteries) before dividing into branches that dive deep into the myocardium (intramural arteries) to supply the cardiac muscle (subendocardial arteries) (Figure 1).


Cardiac muscle
Pericardial space
Epicardial artery
Intramural artery
Left ventricle lumen
Subendocardial artery

QUIZ

Figure 1

We'll next go over the branching pattern of the coronary arteries, which is important because knowing which branch is occluded in atherosclerotic disease will help determine which part of the heart is injured from lack of blood flow.

Parallel Coronary Artery Branches

The two main branch coronary artery (**LCA**) and the **right coronary artery** (**RCA**), as shown in Figure 2A & 2B.

You'll also hear the LCA called the left "main" coronary artery. These two arteries branch independently from the aorta.

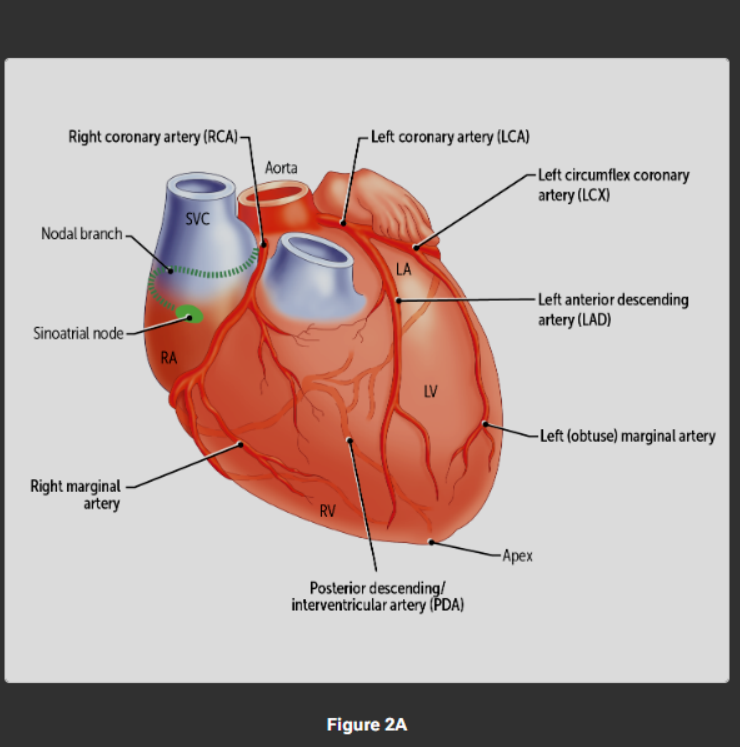

Right coronary artery (RCA)
Aorta
Left coronary artery (LCA)
SVC
Nodal branch
Left circumflex coronary artery (LCX)
LA
Sinoatrial node
RA
Left anterior descending artery (LAD)
LV
Left (obtuse) marginal artery
Right marginal artery
RV
Apex
Posterior descending/ interventricular artery (PDA)

Figure 2A

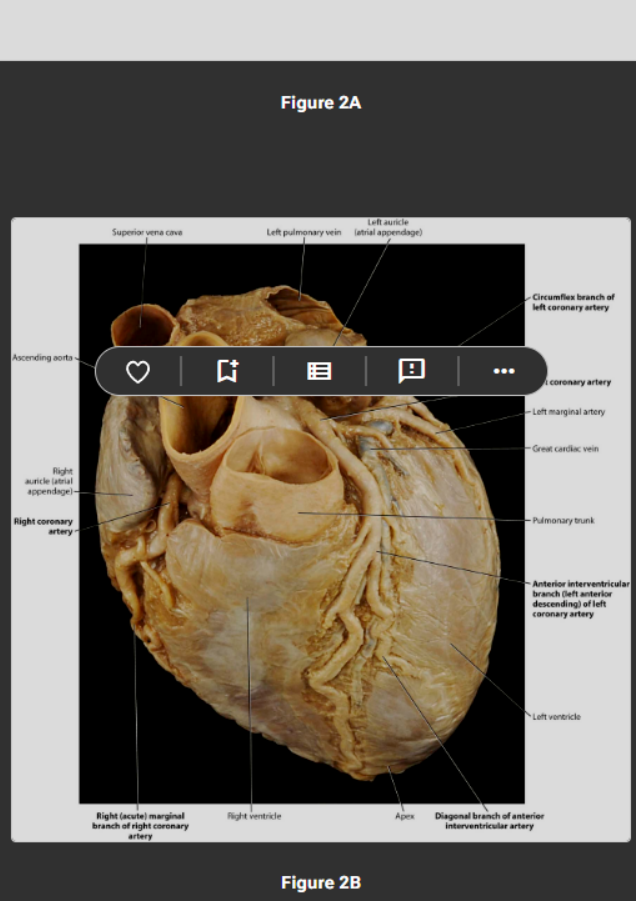

Superior vena cava
Left pulmonary vein
Left auricle (atrial appendage)
Ascending aorta
Circumflex branch of left coronary artery
coronary artery
Left marginal artery
Great cardiac vein
Right auricle (atrial appendage)
Right coronary artery
Pulmonary trunk
Anterior interventricular branch (left anterior descending) of left coronary artery
Left ventricle
Right (acute) marginal branch of right coronary artery
Right ventricle
Apex
Diagonal branch of anterior interventricular artery

Figure 2B

Left (Main) Coronary Artery. The LCA and its branches supply the left atrium (LA) and left ventricle (LV), as well as most of the interventricular septum. Figure 3 shows a coronary artery angiogram, in which dye is injected into the arteries using a catheter that is inserted into a downstream artery (the catheter is the straight vertical structure at left).

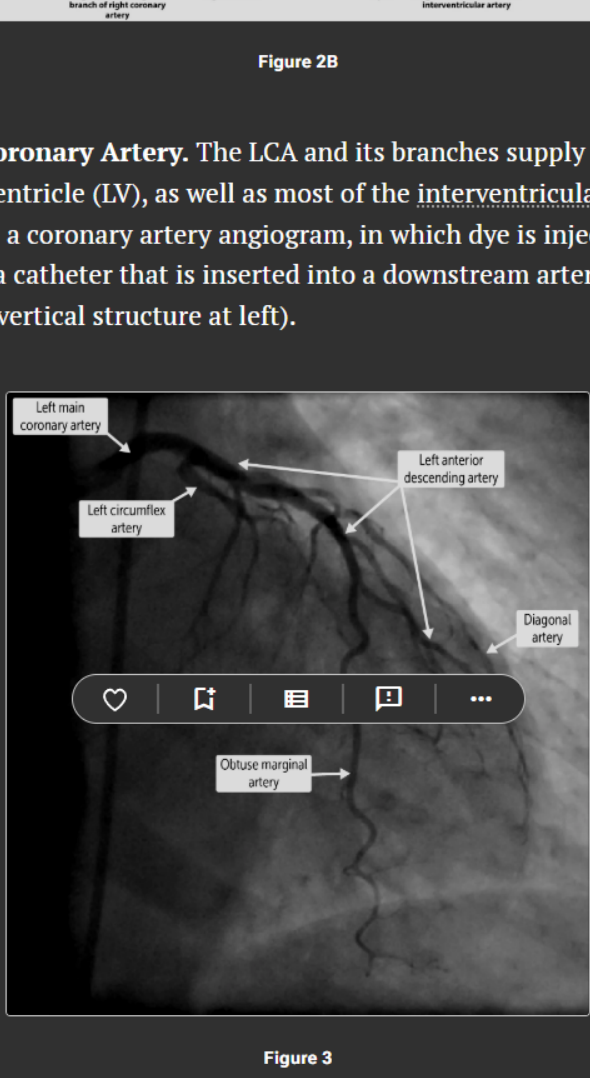

Left main coronary artery
Left circumflex artery
Left anterior descending artery
Diagonal artery
Obtuse marginal artery

Figure 3

You can compare this figure with Figure 1 above to see the relationship to the entire heart. The LCA courses behind the pulmonary trunk and divides into the **left anterior descending (LAD also known as anterior interventricular)** artery and the **left circumflex (LCX) artery**.

The LAD artery travels on the anterior surface of the heart in the anterior interventricular groove to supply the anterolateral papillary muscle, the anterior surface of the LV, and the anterior two-thirds of the interventricular septum (the myocardial wall that divides the LV and right ventricle [RV]). It typically travels all the way to the apex of the heart. The LAD has Diagonal Branches that mostly run diagonally towards the LV (Figure 2B, 3). Therefore, although the LAD provides blood supply to the anterior heart, more is to the LV (which has higher oxygen needs) than the RV.

The LCX travels posteriorly between the LA and LV to supply the LA, and its large, obtuse marginal branch supplies the posterior LV.

Right Coronary Art RCA first passes between the superior vena cava (SVC) and the pulmonary trunk, between the right atrium (RA) and RV, down toward the posterior part of the heart. The sinoatrial (SA) nodal branch begins near the origin of the RCA to supply the SA node (where the heartbeat originates) (Figure 4).

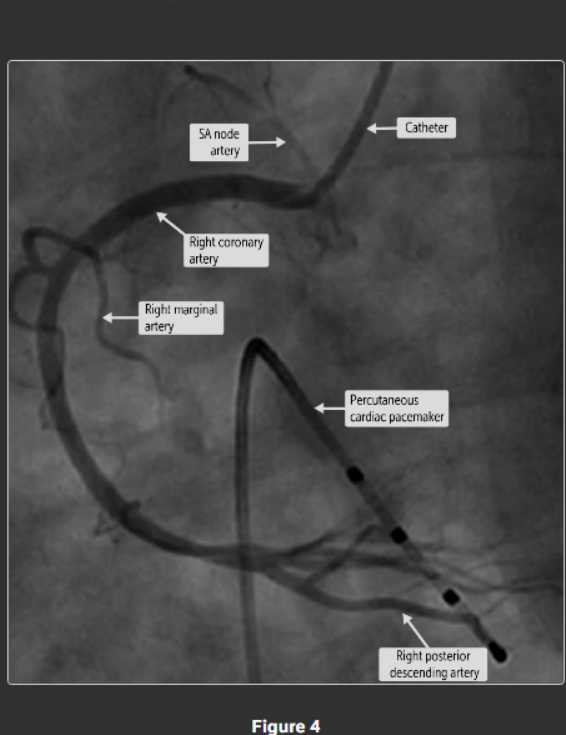

SA node artery
Catheter
Right coronary artery
Right marginal artery
Percutaneous cardiac pacemaker
Right posterior descending artery

Figure 4

Other branches include the right acute marginal artery, which supplies the RV and the apex. The posterior descending/interventricular artery (PDA) originates from the R atrioventricular (AV) node (which regulates the electrical impulse as it travels from the SA node to the ventricles), the posterior one-third of the interventricular septum, the posterior two-thirds of the ventricles, the inferior wall of the heart (diaphragmatic surface), and the posteromedial papillary muscle of the left ventricle.

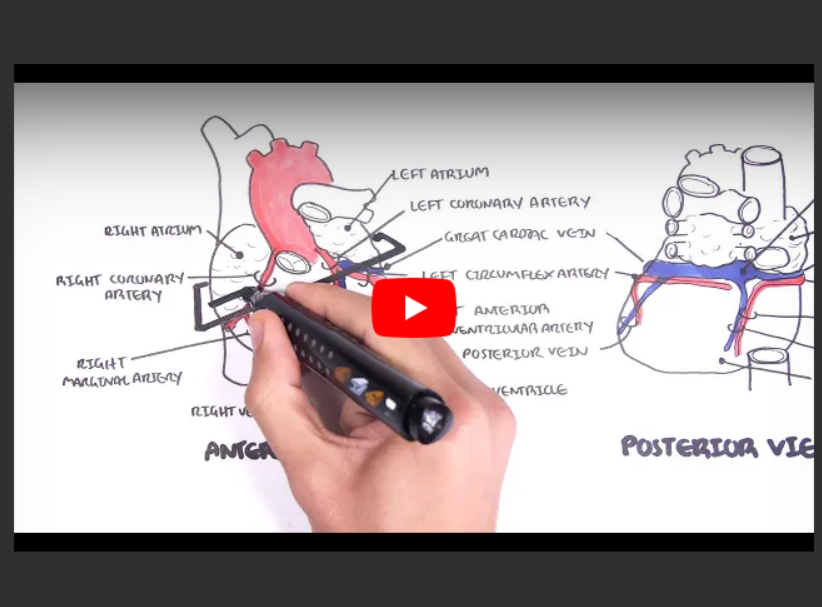
Note that the conduction system of the heart, the Bundle of His and the Right and Left Bundle Branches (discussed in a separate brick), reside in the Interventricular Septum. Occlusion of coronary arteries that run along the Interventricular Septum and perfuse it with septal perforators, the LAD and PDA, can therefore affect this conduction system.

Location of Coronary Arteries Determines their Perfusion Territory and Region at risk of Ischemia from Coronary Artery Occlusion

Artery	Location	Perfusion Territory
Left Anterior Descending	Runs along Anterior IV Septum to the Apex between anterior RV and LV, but the Diagonal Branches are towards the LV	Anterior LV > RV, Apex, Anterior 2/3 of IV Septum with AV Bundles
Diagonals	Branches off LAD towards LV	LV
Left Circumflex	Runs between LA and LV	LA and LV
Left Coronary	LAD + Circumflex	LAD + Circumflex territories
Right Coronary	Runs between RA and RV, near SA Node	RA, RV, and SA Node
Posterior Interventricular posterior descending artery (PDA)	Runs along Posterior IV Septum between RV and LV	Posterior RV, LV, inferior surface, posterior 1/3 of IV Septum with AV Bundles
Right Marginal	margin of RV	Lateral RV
Left Marginal	Branch of Circumflex along lateral margin of LV	Lateral LV

It should be noted that although the RCA is not so involved in the high-energy pumping of the LV, it is critical to keeping the heart's pacemakers and electrical conduction intact.

View this video to have a better understanding of the coronary vessels (draw along with them for maximum learning!)


RIGHT ATRIUM
RIGHT CORONARY ARTERY
RIGHT MARGINAL ARTERY
LEFT ATRIUM
LEFT CORONARY ARTERY
GREAT CARDIAC VEIN
LEFT CIRCUMFLEX ARTERY
POSTERIOR VIEW

What are the main branches of the LCA?

Let's look at some of the typical variations in the arterial structures of the heart. These can be important in coronary artery disease (CAD) to determine which occlusions give rise to which types of damage to the heart.

Variations in the Branching Patterns of Coronary Arteries (LO#2)

First is a variation in artery dominance, meaning which coronary artery gives rise to the posterior descending artery (PDA). Most people (~85%) have a right-dominant coronary artery, meaning that the PDA is a branch of the RCA. The remainder of the population has either left-sided dominance, meaning the LCX gives rise to the PDA, or a codominant circulation, where RCA and LCA combine to give rise to the PDA.

CLINICAL CORRELATION

Patients with right-d after percutaneous coronary intervention (angioplasty), an interventional treatment for myocardial infarction.

Patients with left-dominant circulations have a worse outcome after myocardial infarction. Of the coronary arteries, the left coronary system has a higher rate of coronary artery disease than the right (due to more left coronary blood flow and shear stress, and more oscillatory stress due to stronger left ventricular contractions). In a left-dominant patient, more myocardium is at risk during left coronary system occlusions, whereas a right-dominant patient retains blood flow to the Posterior Descending Artery and perfusion of the posterior Left Ventricle and posterior interventricular septum.

Rarer variations include the following:

- An accessory coronary artery (4%)
- A single coronary artery (which then splits into left and right)
- An LCX that arises from the right aortic sinus
- A variant supply to the SA and AV nodes: the SA and AV nodes are usually supplied by the RCA; however, the LCX's branches may instead supply these structures.

Which c people?

Vessel Blockage and Compensation

Vascular anastomosis is a direct connection between branches of arteries. Coronary arteries are a great example of this. This connection allows blood flow between the vessels, potentially supporting perfusion in areas affected by reduced flow (Figure 5A).

An **anastomosis** can be:

- **Naturally occurring**
- **Surgically created** (e.g., in bypass grafting)
- **Functionally formed** through **collateral vessel development** in response to ischemia

When a vessel becomes blocked, the body very cleverly finds ways to allow blood to flow around a blockage by **collateral circulation** from newly formed blood vessels (angiogenesis) or enlarged blood vessels (arteriogenesis) that reroute blood flow around the obstruction (Figure 5B). This process helps preserve oxygen delivery to heart tissue and can reduce the severity of a heart attack. These vessels may eventually form **functional anastomoses**, meaning they create new connections that allow blood to flow between major vessels.

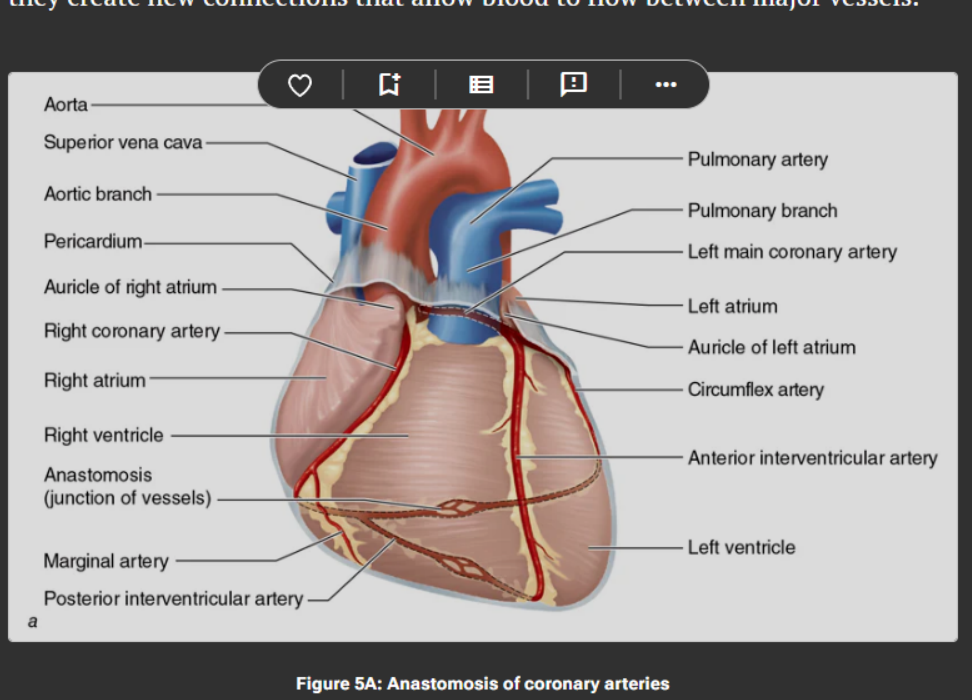

Aorta
Superior vena cava
Aortic branch
Pericardium
Auricle of right atrium
Right coronary artery
Right atrium
Right ventricle
Anastomosis (junction of vessels)
Marginal artery
Posterior interventricular artery
a
Pulmonary artery
Pulmonary branch
Left main coronary artery
Left atrium
Auricle of left atrium
Circumflex artery
Anterior interventricular artery
Left ventricle

Figure 5A: Anastomosis of coronary arteries

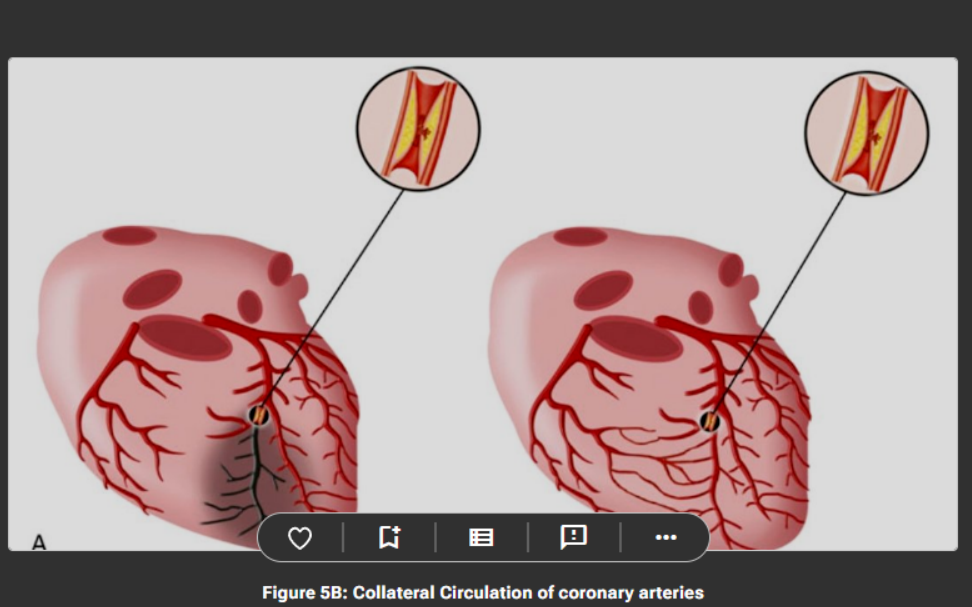


Figure 5B: Collateral Circulation of coronary arteries

Blood flowing through the collateral arteries can bypass the occluded branch of the coronary artery, similar to traffic detours that bypass a blockage on the highway. These collateral vessels can be located between small capillaries or large muscular arteries.

As seen in Figure 5A/B, an anastomosis and collateral circulation can benefit areas of the heart that are prone to ischemia resulting from arterial stenosis or blockage. The anastomosis can shift blood from healthier arterial sources to the stressed region to prevent or lessen a potential ischemia.

Approximately 20% to 25% of healthy patients have arteries that connect the standard coronary arteries, and collateralization increases to 30% to 35% in the population with CAD. Well-developed coronary collateral arteries guard against myocardial infarctions and improve survival. The newly formed anastomoses between vessels typically takes between a few days and a few weeks to form.

How do the heart's vessels try to adjust the blood flow around a blockage in a vessel?

What Are the Primary Cardiac Veins? (LO#3)

Now let's turn to the venous drainage of the heart. There are three primary veins (great, middle, and small) that drain the blood flow from the heart into the **coronary sinus**, where the veins coalesce at the back of the heart within the coronary sulcus (Figure 6). The coronary sinus empties directly into the RA. The superior vena cava, inferior vena cava, and coronary sinus therefore all drain deoxygenated blood into the right atrium, allowing the right ventricle to deliver it to the pulmonary arterial system for oxygenation.

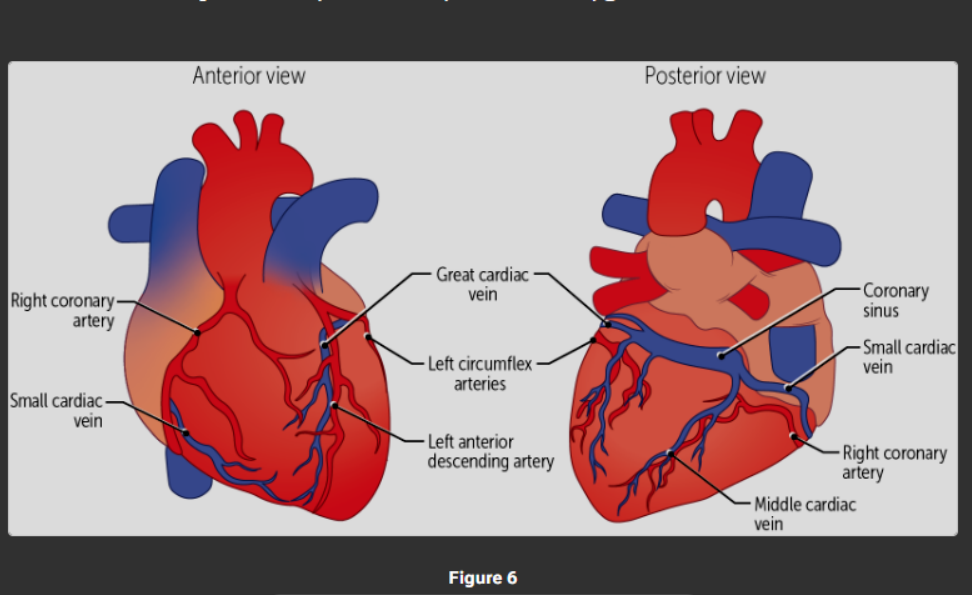

Anterior view
Posterior view
Right coronary artery
Small cardiac vein
Great cardiac vein
Left circumflex arteries
Left anterior descending artery
Coronary sinus
Small cardiac vein
Right coronary artery
Middle cardiac vein

Figure 6

Great Cardiac Vein

This vein starts at the apex of the heart and then runs along the anterior interventricular groove, accompanying the LAD artery. After looping around the LA appendage, it continues in the coronary sulcus alongside the LCX before emptying into the coronary sinus.

Middle Cardiac Vein

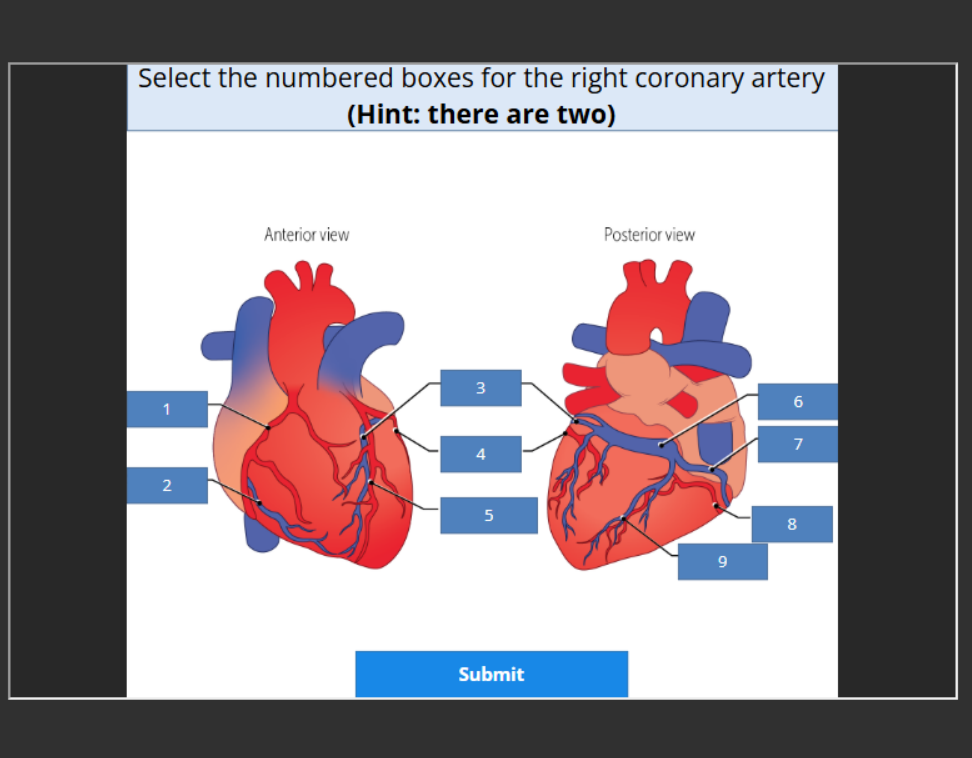
The middle cardiac vein ascends the posterior interventricular groove, running alongside the PDA before draining into the coronary sinus.

Small Cardiac Vein

This vein accompanies the acute marginal artery from the RCA. It then courses in the right posterior AV groove and drains into the coronary sinus.

What are the three primary veins that drain the blood from the heart?

Interact with the knowledge check below to review key vascular structures thus far.


Select the numbered boxes for the right coronary artery
(Hint: there are two)
Anterior view
Posterior view
1
2
3
4
5
6
7
8
9
Submit

What Is the Relationship Between the Lymphatic System and the Heart? (LO#4)

In addition to the coronary vessels that supply and drain the heart, there are lymphatic vessels alo n excess interstitial fluid, pathogens, and cellular products and debris and return them to the venous circulation. The lymphatics also promote optimal healing after myocardial ischemia and infarction.

There are three main lymphatic clusters draining the heart: the subendocardial (below the endocardium), myocardial, and subepicardial (below the epicardium) plexuses. Lymphatic channels leaving the subepicardial plexus form multiple cardiac collecting trunks that drain both ventricles.

These trunks combine with other lymphatic vessels before draining into the **tracheobronchial nodes**. The lymph then enters the thoracic and right lymphatic ducts, and these return the lymph into the venous circulation (Figure 7).

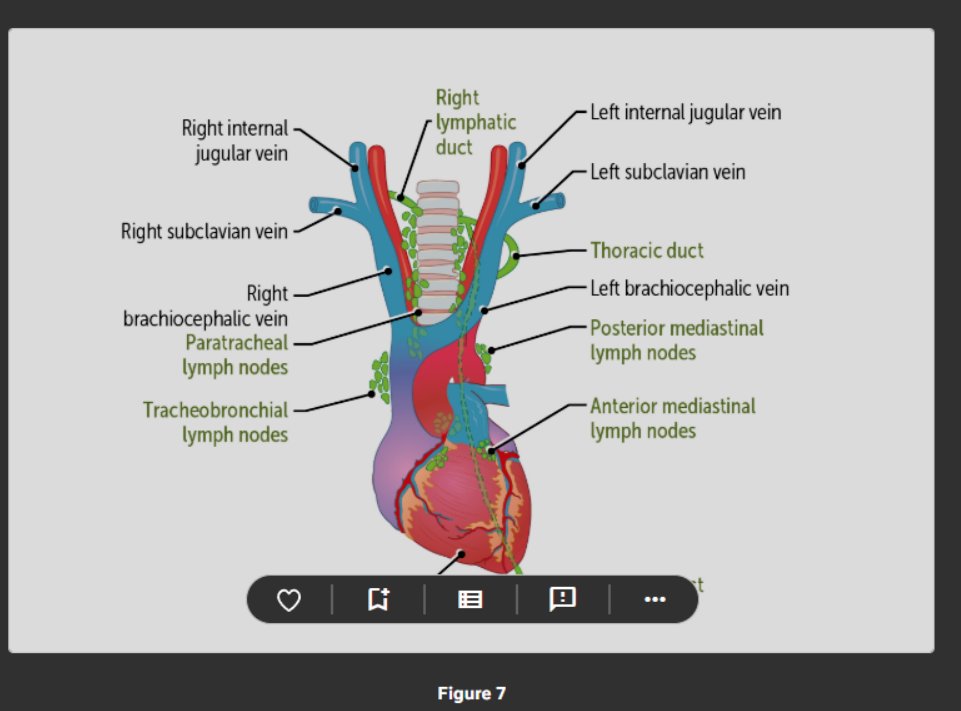

Right internal jugular vein
Right lymphatic duct
Left internal jugular vein
Left subclavian vein
Right subclavian vein
Thoracic duct
Right brachiocephalic vein
Left brachiocephalic vein
Paratracheal lymph nodes
Posterior mediastinal lymph nodes
Tracheobronchial lymph nodes
Anterior mediastinal lymph nodes

Figure 7

What are the main lymph nodes draining the heart?

CASE CONNECTION BACK TO INTRODUCTION ↑

Thinking back to Alex and Sam, why might Sam's outcome have been better?

Everything else being equal, one way to explain the difference in outcomes could be that Sam had well-developed collateral blood flow to the ischemic area or vascular anastomoses, which served to connect the coronary vessels. Both collaterals and anastomoses allow for blood to shift from nonischemic to ischemic areas, lessening infarct size.

Atherosclerosis is m e to increased blood flow, shear stress, and oscillatory stress (from strong LV contractions). Therefore, more specifically, if Alex has a left-dominant coronary system, a left sided coronary occlusion puts more myocardium at risk. Not only are the anterior and lateral left ventricle at risk, but also the Posterior Descending Artery perfusion territory is at risk. If Sam has the more common right-dominant system, his posterior left ventricle and posterior interventricular septum is at less risk. Left-dominant system patients have a worse outcome after acute coronary events.

Summary

- The coronary arteries originate from the aorta and supply the heart with oxygen and nutrients.
- The left coronary artery (LCA) and right coronary artery (RCA) are the main branches, with further subdivisions supplying specific heart regions.
- Coronary artery dominance affects outcomes in coronary artery disease, with most people having a right-dominant system.
- Vascular anastomoses and collateral circulation help maintain blood flow during coronary artery blockages.
- The great, middle, and small cardiac veins drain deoxygenated blood from the heart into the coronary sinus.
- The heart's lymphatic system drains excess fluid and debris, aiding in healing after myocardial ischemia.

Review Questions

1. A 59-year-old male with diabetes and hypertension develops a severe atherosclerotic oc Which of the following structures would be most deprived of oxygen by this occlusion?

A. ○ Interventricular septum
B. ● Left atrium
C. ○ Right atrium
D. ○ Right ventricle
E. ○ Sinoatrial node

Hide Explanation

The correct answer is the left atrium (B). The left circumflex artery (LCX) is a branch of the left coronary artery, traveling posteriorly to supply the left atrium and the left ventricle, so both may become damaged by lack of blood flow in the LCX. The anterior interventricular septum (A) is supplied by the left anterior descending artery. The right atrium (C) is supplied by branches of the right coronary artery (RCA). The right ventricle (D) is mostly supplied by the right marginal branch and the LAD artery. The sinoatrial node (E) is supplied by the sinoatrial nodal branch from the RCA.

2. In most of the population, which coronary artery gives rise to the posterior descending artery?

A. ○ Left anter
B. ○ Left circumflex artery
C. ○ Left coronary artery
D. ● Right coronary artery
E. ○ Right marginal artery

Hide Explanation

The correct answer is right coronary artery (D). Most of the population has a dominant right coronary artery (RCA). This means that the posterior descending artery (PDA) is a most commonly a branch of the RCA. The left anterior descending artery (A) does not usually give branches to the PDA. The left circumflex artery (B) can give rise to the PDA in a smaller proportion of the population, creating a dominant left coronary artery. The left coronary artery (C) does not give branches to the PDA directly. The right marginal artery (E) is not directly associated with the PDA.

3. Which cardiac vein accompanies a branch of the right coronary artery on the anterior aspect of the heart?

A. ○ Coronary sinus
B. ○ Great cardiac vein
C. ○ Middle cardiac vein
D. ● Small cardiac vein

Hide Explanation

The correct answer is small cardiac vein (D). The small cardiac vein accompanies the acute marginal artery from the right coronary artery (RCA). It then courses in the right posterior atrioventricular groove and drains into the coronary sinus. The coronary sinus (A) receives blood from the cardiac veins but is not associated with the RCA. The great cardiac vein (B) runs on the anterior interventricular groove, accompanying the left anterior descending artery. The middle cardiac vein (C) runs on the posterior interventricular groove, accompanying the posterior descending artery.

4. Which of the following lymph nodes drain the heart?

A. ○ Axillary nodes
B. ○ Para-aortic nodes
C. ○ Superficial inguinal nodes
D. ○ Superior mesenteric nodes
E. ● Tracheobronchial nodes

Hide Explanation

The correct answer is the tracheobronchial nodes (E). The subepicardial lymphatic plexus gives rise to multiple left and right cardiac collecting trunks. These eventually drain into the tracheobronchial nodes. None of the other nodes drain the heart. The axillary nodes (A) drain the breast, upper limbs, and skin above the umbilicus. The para-aortic nodes (B) drain the testis or ovaries, uterus, and kidneys. The superficial inguinal nodes (C) drain nearby structures, such as the scrotum or vulva and the lower portion of the anal canal. The superior mesenteric nodes (D) drain the upper part of the bowel (duodenum, jejunum, ileum, and proximal colon).

How would you rate this Brick?

References

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- Parikh NI, Honeycutt EF, Roe MT, et al. Left and codominant coronary artery circulations are associated with higher in-hospital mortality among patients undergoing percutaneous coronary intervention for acute coronary syndromes. Report from the National Cardiovascular Database Cath Percutaneous Coronary Interventi c Qual *Outcomes*. 2012;5:7 coronary intervention for acute coronary syndromes. Report from the National Cardiovascular Database Cath Percutaneous Coronary Intervention (CathPCI) Registry. *Circ Cardiovasc Qual Outcomes*. 2012;5:775-782. doi:10.1161/CIRCOUTCOMES.111.964593
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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Outline the origin, course, and distribution of the coronary arteries.~~
02 ☑ ~~Discuss the typical variations in the arterial structures of the heart.~~
03 ☑ ~~Outline the origin, course, and drainage of the cardiac veins.~~
04 ☑ ~~Describe the lymphatic drainage of the heart.~~

Objective Confidence: 4/4